Does Urban Status Reverse the Poverty-Income Effect on Low Food Access?

DATA471 Individual Project

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1 Executive Summary

This report asks whether the relationship between poverty, income and low food access in US census tracts depends on whether a tract is Urban or Rural. Pooling all tracts together, poverty rate and income barely separate tracts with low food access from the rest of the sample (a weak effect). Splitting by Urban/Rural status, however, shows both relationships pointing in opposite directions in the two area types: in Rural tracts, low-access areas tend to be poorer and lower-income than other Rural tracts, matching common intuition, while in Urban tracts, low-access areas tend to be wealthier and have less poverty than other Urban tracts.

A statistical model that lets poverty and income have different effects in Urban vs. Rural tracts fits significantly better than a model that assumes one fixed effect everywhere ($p < 0.001$). Once poverty and income are modelled together, though, neither one's own effect literally flips sign by area type. Instead, income is the stronger signal of low access in Rural tracts, and poverty is the stronger signal in Urban tracts. Adding a measure of vehicle access to the model did not explain this pattern (its effect did not shrink the way a true explanation would), and the finding for poverty was less stable across subsets of the data than the finding for income.

A random forest model, which can detect patterns without the modeller specifying them in advance, predicted low-access status noticeably better than the statistical model (71% vs. 66% accuracy), and found that vehicle access and tract demographic makeup were stronger predictors overall than poverty or income on their own. The main practical implication: a single poverty or income cut-off is not a reliable way to flag at-risk areas everywhere. Which measure matters depends on whether the area is Urban or Rural, and vehicle access deserves more attention as a screening factor in its own right.

2 Background

Low-income households are often the ones hit hardest by a long trip to the nearest grocery store, since the extra time and transport cost falls on people who can least afford it. That link motivates asking whether poverty and income actually predict low food access.

The Phase 1 group report (Whithair, Busto, Mehmood and Cai, 2026) looked at access to large retail food outlets using the Food Access Research Atlas, and found that two variables often assumed to drive low food access, PovertyRate and MedianFamilyIncome, barely separated low-access tracts from the rest of the sample once the whole dataset was pooled together (a weak

effect by conventional standards), and taken on its own this would suggest poverty and income are not strong predictors of low food access at all.

Splitting the same comparison by Urban/Rural status changes the picture considerably. Within Rural tracts, low-access areas tend to be poorer and lower-income than other Rural tracts, which matches common intuition. Within Urban tracts the pattern reverses: low-access areas tend to have lower poverty and higher income than other Urban tracts. The group report documented the income-side version of this reversal; checking the same comparison for poverty rate shows the same pattern holds there too.

A model that lets poverty rate and income have different effects in Urban and Rural tracts, rather than one fixed slope for the whole country, is the most direct test of whether this reversal is real rather than noise. That gives two linked, testable questions for the individual analysis:

Research question(s). Does a logistic regression model of low food access status (LA1and10) that includes an Urban interaction term for both PovertyRate and MedianFamilyIncome fit significantly better than the same model without the interaction terms? If so, do the Urban-specific slopes for poverty rate and income actually differ between Urban and Rural tracts in a way consistent with the reversal seen in the exploratory analysis?

3 Data Description

The Food Access Research Atlas (FARA) data is a single cross-sectional snapshot from 2019, not a time series: it records one row per US census tract (72,531 tracts across all 50 states), combining binary low-access flags, tract-level population counts, and continuous rates/shares describing each tract's demographic makeup and distance to the nearest large food retailer. Full variable definitions are in the FARA documentation (USDA ERS, 2019; see References).

This analysis uses a subset of that structure, summarised in Table 1. **LA1and10** classifies a tract as low-access at both the 1-mile (Urban) and 10-mile (Rural) distance thresholds, following the "food desert" measurement approach of Ver Ploeg et al. (2009). **PovertyRate** is the percentage of the tract below the poverty line and **MedianFamilyIncome** is the tract's median family income in US dollars (top-coded at $250,001); both are standardised before modelling, and **Urban** (coded 0 for Rural, 1 for Urban tracts) is the interaction variable of interest throughout. **lahunv1share** is the share of housing units in a tract with low vehicle access that are also far from a supermarket, and the four composition shares added in **M3** (Black, children, seniors, and low-vehicle-access households) are each expressed as a proportion of the tract's own population or housing units.

Table 1: Variables used in this report: type, role, and completeness

Variable	Type	Used in	Missing
LA1and10	Binary flag	M0-M3 (response)	None
PovertyRate	Continuous (%)	M0-M3	<1%
MedianFamilyIncome	Continuous ($, top-coded)	M0-M3	<1%
Urban	Binary flag	M0-M3 (interaction)	None
lahunv1share	Continuous (share)	M2 only	27.5% (structural)
Composition shares	Continuous (share)	M3 only	1 tract (negligible)

The core variables (**LA1and10**, **PovertyRate**, **MedianFamilyIncome**, **Urban** and **HUNVFlag**) are close to completely populated: 71,783 of 72,531 tracts (98.97%) have no missing

values on any of them, consistent with the group report's own findings. HUNVFlag is a raw low-vehicle-access flag used here only to define this complete-case sample, not as a predictor. **lahunv1share**'s missingness, by contrast, is structural rather than a data-quality gap: it is missing for 36% of Urban tracts but under 0.1% of Rural tracts, a pattern tied to area type rather than an individual data-quality issue, and it is treated that way in the **M2** analysis below.

One change from the group report's handling of missing data: that report zero-filled missing demographic tract totals (e.g. **TractBlack**) before computing shares, flagged in feedback as misleading since a genuinely missing count isn't the same as a genuinely zero one and treating them alike biases the share downward. The **M3** composition shares below are instead computed as raw ratios (e.g. **TractBlack** / **Pop2010**), left missing wherever an input is missing, and excluded case-wise by any model that uses them.

4 Ethics, Privacy and Security

This section carries over the Phase 1 group report's Ethics, Privacy and Security discussion, revised in response to feedback received on that report: the Security paragraph below now states which measures were actually used, not just which were possible.

Ethics. As in the Phase 1 group report, the clearest ethical risk in this dataset is that correlations involving demographic composition (e.g. the share of a tract's population that is Black) could be read as implying those demographics *cause* low food access, when both are plausibly shaped by the same underlying structural factors (historical redlining, local zoning, retail investment patterns). This risk is sharper here than in the group's EDA, because **M3** below fits a random forest that ranks demographic composition shares by predictive importance, and a ranked importance table is easy to misread as a causal ranking. Results in this report are presented as associations useful for predicting which tracts are low-access, not as evidence about what *causes* low access, and that distinction is kept explicit wherever the random forest's feature importance is discussed.

Privacy. Every variable used is a tract-level count, rate, or flag; there are no individual- or household-level identifiers anywhere in the dataset, so there is no realistic risk of re-identifying a specific person. A small number of tracts have very low population counts and could in principle be distinguishable; none of the analyses here report results for individual small tracts, only aggregate model coefficients and bulk tract-level predictions, so this risk does not apply in practice.

Security. FARA data is public and needs no additional security measures beyond normal good practice. These are the actual measures used for this project, not hypothetical ones: the code and this report are kept in a private GitHub repository during development, and the only device holding a local copy is password-protected. With more time, access to that repository could additionally be restricted with two-factor authentication on the GitHub account, though this was not set up for the project.

5 Exploratory Data Analysis

This section shows only the comparisons that motivate the Detailed Analysis, rather than repeating the group report's full EDA. Figure 1 shows PovertyRate (top) and MedianFamilyIncome (bottom) by LA1and10 status, compared three ways: pooled across all tracts, then split into Rural and Urban. Pooled, the two groups overlap heavily on both variables, matching Background's description of a weak pooled effect. Splitting by Urban/Rural status, however, uncovers the reversal described in Background: within Rural tracts the pattern matches common intuition, while within Urban tracts it reverses. This is the entire basis for the research question above: a relationship that flips sign between two subgroups cannot be captured by a model that only

allows one fixed slope for the whole country, which is exactly what the Urban interaction terms in **M1** are for.

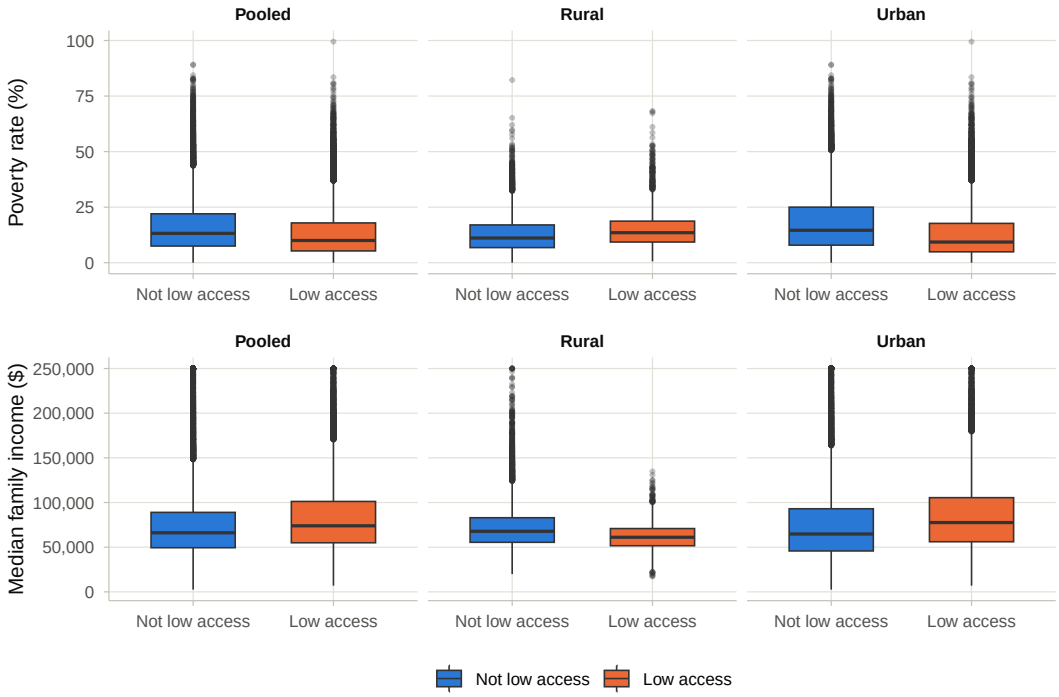


Figure 1: PovertyRate (top) and MedianFamilyIncome (bottom) by LA1and10 status: pooled, then split by Urban/Rural.

A second relationship worth checking before modelling is between the two predictors themselves. Figure 2 shows PovertyRate and MedianFamilyIncome are moderately correlated across all 71,783 tracts (r = -0.66): poorer tracts tend to have lower median income, as expected. This matters later in Detailed Analysis Results, where it is the reason the two predictors cannot be treated as fully independent effects, and the reason PovertyRate and MedianFamilyIncome occasionally point in different directions when the parametric **M1** model is compared against the non-parametric random forest.

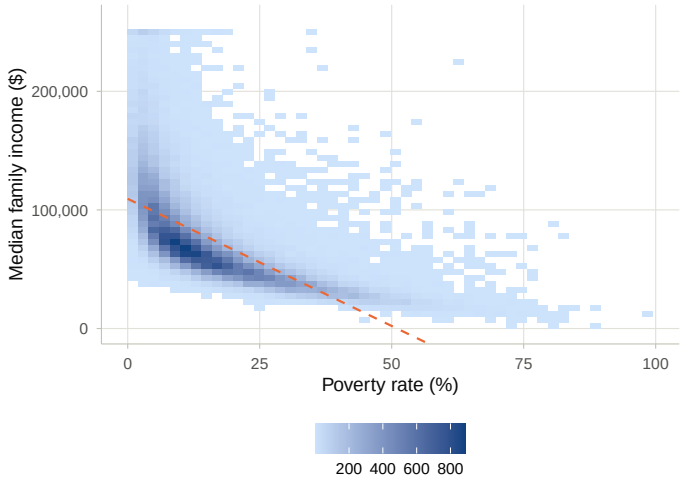


Figure 2: PovertyRate vs. MedianFamilyIncome across all 71,783 tracts (binned density; r = -0.66).

A third variable, explored in detail later in **M2**, is lahunv1share: the share of a tract's housing units with low vehicle access that are also far from a supermarket. Figure 3 shows it by LA1and10 status, split by Urban/Rural, for the tracts where it is defined (its missingness is structural; see Data Description). In both area types, low-access tracts have a higher vehicle-access share than other tracts, consistent with common intuition, but the gap is far larger in Urban tracts (0.07% to 1.51%) than in Rural tracts (2.57% to 3.27%). That makes vehicle access a plausible candidate to explain the Urban-dependent poverty/income reversal, a possibility tested formally in **M2**. **M3** below uses a second, separately-computed vehicle-access measure (hunv_share_plain) with different missingness, not this one; see its caveats for why the two are not a straightforward replication of each other.

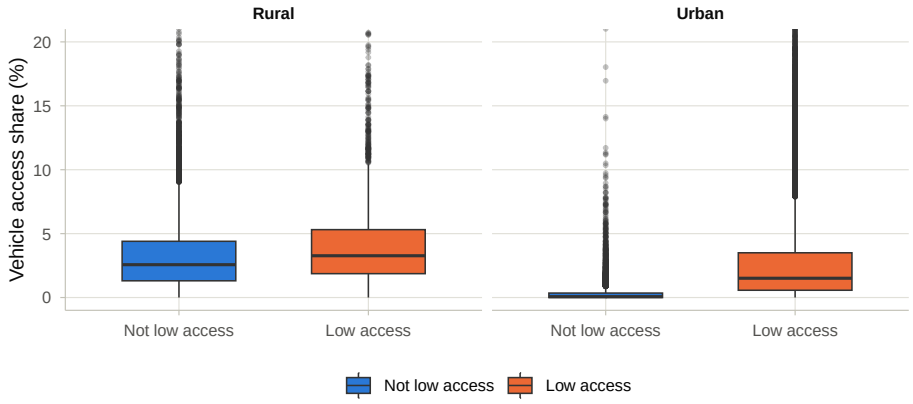


Figure 3: lahunv1share (vehicle access share, %) by LA1and10 status, split by Urban/Rural. Y-axis zoomed to the 0-20% range (99th percentile = 19%); whiskers and outliers are computed on the full data.

6 Detailed Analysis Results

6.1 Method

Logistic regression suits this question well: LA1and10 is binary, the two predictors of interest are continuous, and an interaction term is the right tool for directly testing whether a slope changes between two groups (Urban/Rural) rather than eyeballing two separate box plots. Two models are compared: **M0**, with PovertyRate_z, MedianFamilyIncome_z and Urban_f as main effects only, and **M1**, which adds an Urban interaction term for both continuous predictors. A formal test comparing the two models is the test for whether the Urban-dependent reversal seen in the EDA is real, rather than assuming a single shared slope that happens to look weak overall.

6.2 Data preparation for this analysis

PovertyRate and MedianFamilyIncome are standardised (z-scored) before being used as predictors, so odds ratios are interpretable as "per one standard deviation" and the two variables, which are on very different raw scales (0-100 versus tens of thousands of dollars), are directly comparable within the model. Urban is coded as a two-level factor (Rural/Urban). The modelling dataset is the same complete-case sample described in Data Description (71,783 tracts); no imputation is used for **M0**/**M1**. Missingness on these core variables is small (1.03%) but is concentrated in very small-population tracts (median Pop2010 1,397 among missing rows vs. 4,027 among complete rows), consistent with Census suppression of unreliable estimates for small samples; dropping these tracts has a negligible effect on the modelling sample. **M2** and **M3** use additional variables with their own missingness patterns, handled separately in those sections.

6.3 Model specification

M0: `LA1and10 ~ PovertyRate_z + MedianFamilyIncome_z + Urban_f`

M1: `LA1and10 ~ PovertyRate_z * Urban_f + MedianFamilyIncome_z * Urban_f`

Both are fitted by maximum likelihood (binomial family, logit link) on the full n = 71,783 complete-case sample.

6.4 Results (M0 vs M1)

The formal test comparing **M0** to **M1** is strongly significant ($p < 0.001$), so the data reject the null hypothesis that PovertyRate and MedianFamilyIncome have the same relationship with low food access in Urban and Rural tracts. **M1** also fits the data better overall (Table 2), though the improvement is modest in absolute terms, as expected for tract-level demographic predictors of a binary outcome.

Table 2: Model comparison: M0 vs M1

Model	AIC	McFadden pseudo-R2
M0: main effects only	89551.9	0.061
M1: with Urban interaction	88397.0	0.073

Poverty rate and income are not so closely linked that this becomes a problem (VIF = 1.83 for both, consistent with Figure 2's r = -0.66), so both were kept in the model. Table 3 and Figure 4 break the Urban-specific slopes down by predictor and area (that slope for each predictor is the main-effect coefficient *times* the interaction odds ratio, not the interaction coefficient alone):

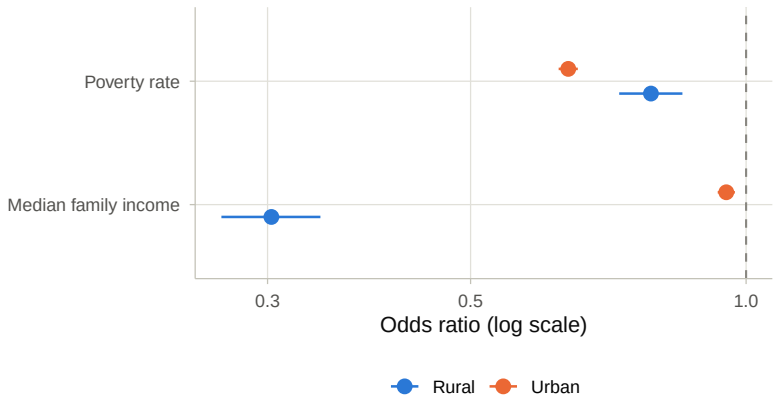


Figure 4: M1 odds ratios (per 1-SD) by Urban status, with 95% CIs.

Table 3: M1 (with Urban interaction): odds ratios and 95% CIs

Term	Odds Ratio	Lower CI	Upper CI	p-value
(Intercept)	0.156	0.146	0.166	<0.001
Poverty rate (z)	0.787	0.727	0.852	<0.001
Urban	5.127	4.809	5.474	<0.001
Median family income (z)	0.303	0.267	0.343	<0.001
Poverty x Urban	0.812	0.748	0.883	<0.001
Income x Urban	3.141	2.771	3.568	<0.001

Income dominates in Rural tracts and poverty dominates in Urban tracts, with each predictor's weaker effect nearly nulled out in the other area type – a more specific finding than the raw EDA boxplots suggest. Considered one predictor at a time (Figure 1), both poverty and income appear to flip sign between Urban and Rural, the same reversal already named in Background. Modelled together, though, neither predictor's own net effect actually changes sign by Urban status: both stay in the same direction (higher poverty/income linked to lower odds of low access) in both area types – only which predictor dominates changes. Because the two predictors are correlated (Figure 2), the apparent bivariate reversal for each one considered alone is, at least in part, a product of shared variance redistributed differently by area type, not two independent reversals.

6.5 Goodness of fit (chosen model: M1)

A formal test of whether **M1**'s predicted probabilities match the data (Hosmer, Lemeshow and Sturdivant, 2013) rejects good calibration (X-squared = 634.18, df = 8, $p < 0.001$) – expected at this sample size (n ~ 72,000), since the test's power to detect any deviation, however small, grows with n. The calibration plot (Figure 5) shows most deciles tracking the diagonal closely, but the top decile under-predicts risk by about ten percentage points (55% predicted vs. 65% observed) – a gap worth flagging for the screening use proposed in Conclusions and Recommendations, since it falls exactly where that use case cares most:

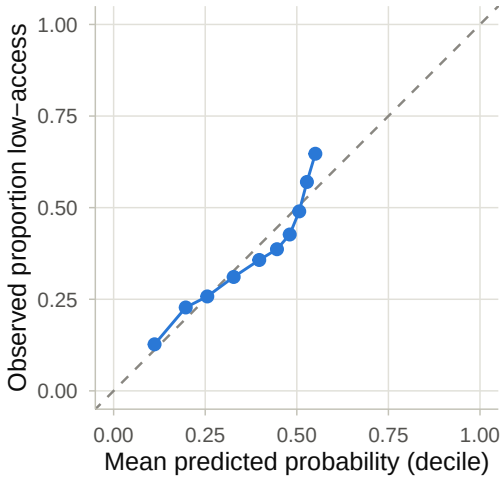


Figure 5: Calibration plot for **M1**: predicted vs. observed low-access rate by decile of predicted probability.

6.6 M2: does vehicle access explain the reversal?

M2 checks an optional sensitivity question from the oral assessment: does vehicle access (lahunv1share, structural missingness; see Data Description) explain the Urban-dependent reversal? It does not. **M2** adds lahunv1share to **M1** as a plain main effect, not interacted with Urban, since the question is whether it explains the existing Urban interaction, not whether it has its own Urban-specific slope.

Restricting to the subsample with vehicle-access data (72.5% of the full sample) already changes the conclusion for poverty: the Poverty x Urban interaction, significant in the full-sample **M1** (Table 3), is no longer distinguishable from zero here – unlike the income-side result, which stays significant under the same restriction (full estimates in Appendix A, Table A1). Adding lahunv1share itself does not support a simple "vehicle access explains it" story: both Urban interaction coefficients should shrink toward zero if it did, but instead income barely moves and poverty grows back to significant – the opposite.

The more defensible reading: vehicle access correlates with both poverty and the subsample's

composition, making the subsample-only estimate unstable – not that it causally explains the reversal. Vehicle access remains worth investigating further (as Dr McMillan suggested), but this analysis does not show it to be the mechanism behind either reversal.

6.7 M3: random forest comparison

A second, non-parametric method (random forest, Breiman, 2001) is worth fitting for two reasons: it can represent any Urban interaction without one being specified in advance, directly checking whether **M1**'s interaction-term story survives a method with no fixed functional form; and it satisfies the brief's suggestion to fit an actual prediction/ML model. It uses the same three predictors as **M1** (PovertyRate_z, MedianFamilyIncome_z, Urban_f) plus four demographic composition shares (Black, children, seniors, low-vehicle-access), all seven included directly since the brief does not call for a feature-selection step.

The final random forest (300 trees) is trained on a 70% split of the data, stratified jointly by LA1and10 and Urban status, and evaluated on the remaining 30%.

6.8 Results (M3)

Partial dependence vs. M1. To check whether the Urban-dependent pattern found in **M1** reflects a real feature of the data, not an artifact of the regression's parametric form, Figure 6 shows how the random forest's own predicted probability changes as PovertyRate and MedianFamilyIncome vary by Urban/Rural status, without the forest ever being told these predictors interact with Urban status.

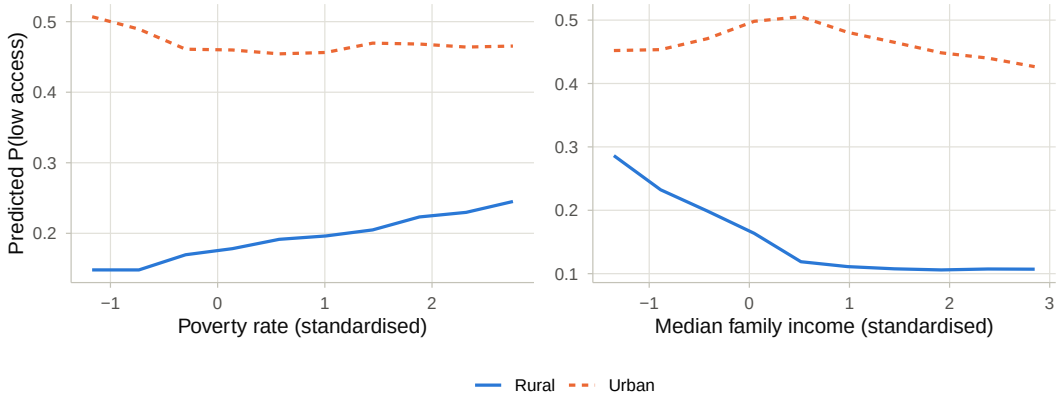


Figure 6: Partial dependence: PovertyRate (left), MedianFamilyIncome (right), by Urban/Rural.

Table 4: Random forest's partial-dependence direction vs. M1's adjusted odds ratio, by predictor and area

Predictor	Area	Forest	M1 OR	M1	Agree
Poverty rate	Rural	Rises	0.787	Falls	No
Poverty rate	Urban	Falls	0.639	Falls	Yes
Median family income	Rural	Falls	0.303	Falls	Yes
Median family income	Urban	Falls	0.951	Falls	Yes

Table 4 checks each of the four curves' direction against **M1**'s adjusted odds ratio. Three of four agree; only Rural poverty does not – the forest's predicted probability *rises* with poverty rate there, tracking the raw bivariate relationship (Figure 1) rather than **M1**'s adjusted Rural

coefficient, which is negative net of income. This disagreement is informative, not a problem: **M1**'s poverty coefficient holds income fixed, while the forest's curve lets income vary naturally alongside poverty, the way the two actually move together (Figure 2) – the two methods answer different questions, so they need not agree, and for Rural tracts here they do not.

Broader pattern scan. A broader scan of the other four **M3** predictors, requested at the oral assessment, confirms the Urban-dependent pattern is not unique to poverty and income (same procedure as Figure 6; full results in Appendix A, Table A2). Three of six show a genuine Rural/Urban direction flip: PovertyRate (above), % children, and, by far the largest effect, the share of low-vehicle-access households, where Rural barely moves but Urban swings sharply – plausible given dense, walkable Urban tracts combine high car-free shares with good food access, a link that doesn't hold where Rural distances are large regardless. The other three (income, % Black, % Seniors) differ in strength but not direction.

Predictive performance. Table 5 shows **M3**'s non-overlapping, higher AUC and accuracy on the same held-out tracts, making it a meaningfully better predictor than **M1**. This does not contradict **M1**'s result: **M1** tests a specific hypothesis about how two slopes vary by Urban status, while **M3** is free to use any combination of features and non-linear interactions, including ones **M1** never specified.

Table 5: Held-out test performance: M1 vs. M3 (bootstrap 95% CI on AUC)

Model	AUC	95% CI	Test Accuracy	Test n
M1 (logistic)	0.686	(0.679, 0.693)	0.662	21536
M3 (random forest)	0.776	(0.769, 0.782)	0.711	21536

Variable importance. Figure 7's feature importance ranking puts vehicle access (hunv_share_plain_z) highest and Urban_f lowest of all seven – not because Urban is unimportant, but because its contribution in **M1** is entirely through interaction terms that a plain importance ranking cannot decompose.

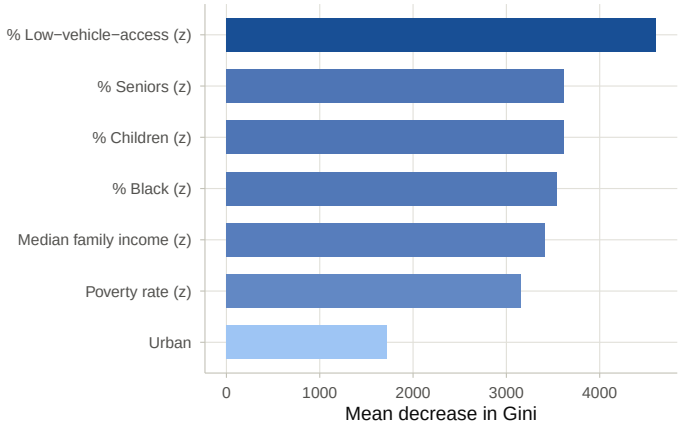


Figure 7: Random forest variable importance (mean decrease in Gini).

Caveats. Two caveats apply to **M1** and **M3** alike, and a third is specific to comparing **M2** and **M3**. The data are aggregated at the census-tract level, so any relationship found here is a tract-level (ecological) association that need not hold for individual households. MedianFamilyIncome is top-coded at $250,001, compressing the top of the income distribution and likely understating the true income effect in the wealthiest tracts. Separately, **M2**'s finding of structural, non-random missingness in lahunv1share does not apply to **M3**: hunv_share_plain and the other composition

shares are defined for all but a single tract, a negligible gap by comparison – a reminder that vehicle access is measured two different ways across **M2** and **M3** (see EDA), so the two results are not a straightforward replication of each other.

7 Conclusions and Recommendations

The Urban interaction in **M1** is real and well-supported ($p < 0.001$), but the mechanism is more specific than a simple sign reversal: income dominates the poverty-income relationship with low food access in Rural tracts, and poverty dominates it in Urban tracts, rather than either predictor's net effect flipping sign once the other is controlled for. **M2** finds no evidence that vehicle access explains this pattern, though the poverty-side finding is less robust to which subset of tracts is analysed than the income-side finding is. **M3**'s random forest confirms the interaction model's predictors are useful for prediction (AUC 0.776 vs. 0.686 for **M1**), while showing that vehicle access and demographic composition shares carry more raw predictive weight than poverty and income once non-linear interactions are allowed. A broader scan of the random forest's own predictions confirms Dr McMillan's oral-assessment suggestion that this isn't unique to poverty and income: three of six predictors scanned (poverty, % children, and above all vehicle access) show a genuine Rural/Urban direction flip, with vehicle access the largest reversal found.

Recommendation. Tract-level screening should not assume a single poverty or income threshold works the same way in Urban and Rural areas. Which predictor signals "at risk of low access" depends on area type, and vehicle access, not just income or poverty, is worth its own screening variable given its importance in **M3**. **M1**'s calibration is weakest exactly in the highest-risk decile (Goodness of fit), so any screening threshold should be validated against observed outcomes rather than applied directly from the fitted probabilities.

Limitations. All three **M3** caveats apply overall (tract-level aggregation, income top-coding, the two vehicle-access measures across **M2** and **M3**). The analysis is also observational and cross-sectional (Data Description): none of **M1**, **M2** or **M3** can distinguish a causal mechanism from a stable correlation, and the Ethics section's caution about demographic composition shares applies equally here. Future work could add a variable-importance stability check to **M3**, or revisit **M2** with a wider range of vehicle-access variables now that **M3** shows vehicle access is the most predictive feature available.

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Appendix A: Supporting detail

Full detail for two secondary checks summarised in Detailed Analysis Results: the **M2** sensitivity analysis ("**M2**: does vehicle access explain the reversal?") and the broader-pattern scan beyond poverty and income (Results (**M3**)).

Table A1. Full coefficients for the **M2** sensitivity check.

Model	Term	Estimate	Std. Error	p-value
M1 (refit on M2 subsample)	Poverty x Urban	0.013	0.043	0.762
M1 (refit on M2 subsample)	Income x Urban	1.261	0.065	<0.001
M2 (+ lahunv1share)	Poverty x Urban	0.534	0.051	<0.001
M2 (+ lahunv1share)	Income x Urban	1.239	0.070	<0.001

Table A2. Full broader-pattern scan: random forest's predicted probability, Rural vs. Urban net change across each predictor's observed range.

Predictor	Rural	Urban	Pattern
Poverty rate	0.097	-0.042	Direction differs
Median family income	-0.179	-0.025	Same direction
% Black	-0.023	-0.075	Same direction
% Children	-0.050	0.163	Direction differs
% Seniors	0.079	0.169	Same direction
% Low-vehicle-access households	0.024	-0.554	Direction differs

Appendix B: Declaration of AI Use

Claude Code was used to help check and improve the R code for the statistical and random forest models, to independently verify numerical claims in the report against the underlying computations, to act as a sounding board for discussing analysis and structural decisions, to assist with data visualisation design, and to assist with Quarto/LaTeX typesetting and English language polishing. All written interpretations and analytical decisions are my own work.